Chapter 8 AI-Driven Management and Modeling Decision Optimization as a Timely Opportunity at the US Department of Defense

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8.1 Why Act Now and Why Engineering Lifecycle and AI?

8.1.1 The Global Driver for AI

The aerospace and defense sector stands at a pivotal moment, necessitating a significant boost in speed and efficiency to sustain global defense superiority. The US DoD and its prime contractors navigate a highly regulated, politically influenced landscape characterized by intricate budgetary, acquisition, standards, and oversight frameworks. This antiquated system, nearly a century old, hinders the requisite acceleration to fulfill the DoD's mission of perpetual victory and troop safety, leading to system delivery delays of 70–100%. This predicament is further compounded by the growing AI requirements and knowledge gaps within the defense sector. Figure 8.1 depicts the challenge as presented by many senior commanders at industry day events in 2023 and 2024.

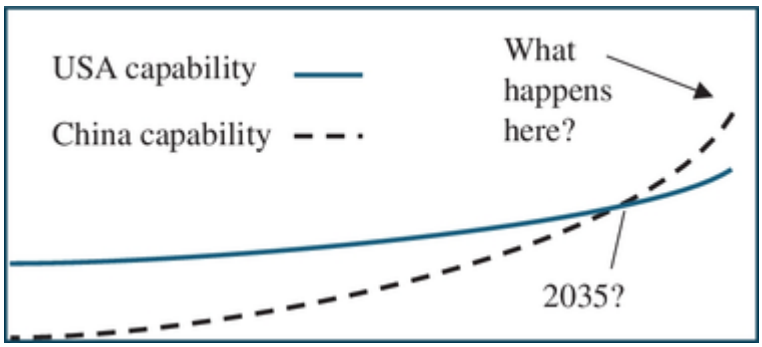


Figure 8.1 United States–China Projected military capabilities to 2035 presented by Leadership in 2024.

Despite multiple endeavors to refine processes and tools for planning, design, and sustainment, the improvements have been minimal. A substantial leap in program and engineering management maturity is imperative, especially during program inception, acquisition, development, production, and sustainment phases. However, a recent regression in systems engineering maturity has exacerbated delays, regressing many programs to inefficient paper-based and office document workflows. The potential of point tools and lightweight platforms remains largely unfulfilled, contributing to consistently poor performance evaluations by the general accounting office.

Artificial intelligence (AI) presents substantial opportunities for the DoD, promising enhanced quality in engineering work products, decision-making, and compliance. This can be achieved through generative task products, predictive analytics, safety/hazard analysis, concurrent simulation and training program integration, and program change management. By incorporating AI management into the lifecycle, leveraging machine learning (ML), deep learning (DL), and generative AI, the existing workforce can surmount the obstacles posed by traditional delivery methods and catalyze rapid advancements in program and product delivery. Addressing data quality and access issues are equally vital steps toward realizing this vision. These advantages are particularly timely given the heightened focus on efficiency and competitiveness on both national and international stages.

8.1.2 Urgency

This section underscores the critical need for a comprehensive AI ecosystem management and modeling approach for military engineers, DoD executives, and military commanders. It elucidates the current predicament, the urgency for transformation, and the envisioned solution to uphold the US defense superiority.

The defense sector is undergoing an unprecedented transformation, propelled by rapid technological advancements and shifting geopolitical threats. To preserve superiority, particularly within the US DoD, the integration of AI and advanced decision-support tools into intricate systems engineering is indispensable. This chapter explores how harnessing an AI ecosystem management and modeling capability, underpinned by AI, can optimize decision-making across all stages of the defense program lifecycle, from design to deployment.

The US DoD currently operates in a complex environment where the balance of power is shifting due to adversaries' technological acceleration. In the next decade, the United States risks losing its edge unless it undergoes a radical transformation of its defense program processes. AI presents the potential for "decision optimization at every desk" within an integrated and dynamic ecosystem. This is not merely a matter of competitive ad-

vantage, but a necessary evolution to guarantee mission success and strategic dominance on the global stage.

The aerospace and defense industries are mired in processes that are bureaucratic and slow, resulting in substantial delays in the delivery of critical systems. These delays, often 70–100% longer than anticipated, are primarily due to an outdated acquisition process and inadequate adoption of AI-driven tools. The industry's persistent reliance on conventional methods further compounds these inefficiencies, making it imperative for the DoD to transition from paper-based workflows to dynamic, data-driven systems powered by AI.

8.1.3 No Math Gets Us "There" with What We Have Today

The current defense engineering and acquisition systems are designed to yield inefficiency, characterized by siloed data, suboptimal decision-making, and a lack of real-time insights. AI, however, offers the potential to automate routine tasks, enhance data precision, and introduce predictive capabilities that could significantly expedite delivery timelines. W. Edwards Deming's assertion that "96% of problems are systems-driven, and only 4% are people-driven" underscores the need for a comprehensive overhaul of the DoD's antiquated systems.

Recent modernization efforts have made some headway in addressing "technical debt" through platform implementations and modeling, but adoption rates remain low, and the potential for substantial speed and precision improvements is minimal. This chapter proposes a unified approach to managing DoD programs, potentially the entire enterprise, at both government and prime contractor/lead systems integrator levels. An AI ecosystem management and modeling approach is advocated, detailing a new ecosystem encompassing people, process, platform, and information (PPTI) essential for achieving superiority through faster, better, cheaper, and smarter delivery.

This value will be realized in mission capability and kill chains/value chains, in conjunction with dependent programs, where operational capabilities and services collaborate effectively. These architectures must be

meticulously planned, designed, and maintained within an AI ecosystem management and modeling ecosystem, aiming for decision optimization at every desk, based on real-time data.

AI can augment decision support and optimization by processing vast volumes of data to generate actionable insights. In defense programs, decision-making must be optimized across all hierarchical levels – from executive leadership to technical chiefs and program managers. Current decision-support systems, fall short of what AI-enhanced systems can achieve. Implementing AI-driven tools across the ecosystem would not only improve data accuracy but also enable more responsive and informed decision-making at every operational level.

8.1.4 DoD Program-level Challenges

The DoD's programs have become increasingly complex, making it challenging for the government to effectively plan, design, operate, and manage critical programs for timely field delivery. Significant changes are urgently required. Currently, many DoD programs rely on basic office document exchanges, shared drives, shared repositories, email attachments, and monolithic point tools, which often obstruct data sharing rather than facilitate it. Systems engineering efforts are typically embedded within these document-intensive environments. Frustration with point tools has led to their disregard or elimination, with a return to office documents, spreadsheets, and presentation decks. Spreadsheets, limited to two or three dimensions, fail to capture the multiple dimensions of linked data necessary for program elements. This approach has proven inadequate, worsening delays in speed-to-mission relative to adversaries.

In addressing these challenges, it is insightful to look at how other industries and agencies have managed similar complexities. Two notable examples are NASA's strategic outsourcing and the automotive industry's lifecycle integration.

NASA's strategic outsourcing strategy involves partnering with private companies to develop and operate spacecraft and other critical systems. This approach has allowed NASA to leverage the expertise and resources

of the private sector, leading to more efficient and cost-effective program execution.

The automotive industry has also successfully integrated its lifecycle, from design and development to manufacturing and maintenance. This integration has enabled industry to streamline its processes, reduce costs, and improve product quality.

These examples demonstrate the potential benefits of adopting more integrated and efficient approaches to managing complex programs. By learning from these industries, the DoD can identify the best practices and strategies that can be adapted to its unique context.

8.1.5 Realization and Subsequent Brave Decision by NASA to Outsource Low Earth Orbit Launch

The defense sector must draw inspiration from industries like NASA and the automotive sector, which have successfully utilized AI and lifecycle integration to enhance operational efficiency. For instance, NASA's decision to outsource low Earth orbit (LEO) launch operations to private industry allowed it to concentrate on higher-priority missions. Similarly, the automotive industry's adoption of model-based systems engineering (MBSE) reduced development cycles by 50%. These case studies suggest that systems engineering if properly integrated, could significantly reduce time-to-field for defense systems while improving decision-making fidelity. Why not add systems engineering methods for AI delivery?

8.1.6 The Automotive Industry

The automotive industry, facing survival challenges, adopted an integrated lifecycle platform and MBSE approach. This strategic overhaul resulted in $500 million in savings per new car development and halved the lifecycle from idea to production to 5 years, compared to the previous 10 years. This integration also facilitated better management of global parts for common platforms.

8.1.7 What Are the Consequences of Inaction?

If the DoD fails to act decisively, it risks becoming increasingly irrelevant in critical areas such as AI Autonomy. The current acquisition and delivery system is inadequate for achieving the required delivery speeds necessary to maintain competitiveness and dominance. Significant new programs must be delivered within 5–10 years, despite challenges such as the retiring workforce, poor business and data systems, and difficulties in attracting new talent. To overcome these obstacles and ensure timely and effective program delivery, a modern lifecycle automation approach is essential.

8.1.8 Culture, Mindset, or Perspective?

Decision-making within DoD programs often relies on data at the desk that is less than 20% accurate, on average, resulting in cascading inefficiencies as programs progress. While leadership frequently emphasizes the need for cultural change, a top-down directive alone is insufficient. Instead, improvements in processes, tools, and engineering techniques at the desk level are needed to drive meaningful change. Cultural shifts will follow the demonstrated value of these enhancements, not precede them. Therefore, the focus should be on changing perspectives and procedures to foster a high-precision decision-making environment.

8.1.9 Beyond the United States

Recent events in Europe and the Southern Pacific underscore the necessity for integrated warfighting capabilities across the US services, DoD agencies, NATO, and other allies. Differing doctrines, texts, and training methods complicate interoperability. Looking ahead, interchangeability – a concept allowing various equipment packages to be used across different platforms (and nations), such as the US missiles on MIGs – will be critical. Achieving this capability is highly improbable with the current approach to program engineering, interoperability, and ecosystems. An AI ecosystem management and modeling approach is necessary to support these goals.

8.2 Who Needs to Make Changes in the Ecosystem?

Achieving dramatically higher speed and precision necessitates the active participation of diverse stakeholders across the program ecosystem. This includes executives, senior management, technical chiefs, program managers, and a multitude of other key players. Their collective involvement is crucial for driving the advancements needed to meet the demands of modern defense initiatives. The ever-evolving landscape of laws, regulations, policies, and guidelines necessitates robust and efficient compliance capabilities within the DoD lifecycle. An integrated governance regime and AI capabilities embedded within a unified lifecycle platform can streamline this process, ensuring swift and successful satisfactory delivery to the warfighter. This paper elucidates the significance of AI in enhancing compliance, predictive analytics, and workforce development. AI can provide invaluable insights from diverse data sources throughout the lifecycle, properly. Trained AI models can analyze user data, operational data, platform data, and more, generating predictions and prescriptive actions.

Both traditional Deep Learning (DL) models and Large/Small Language models play a pivotal role in generating, analyzing, and optimizing critical program data like requirements, as-is and to-be architecture, change management, program/portfolio management, and test management. Continuous access to compliance documents, specifications, and program documents, can be analyzed “live” at the desk or in the field by extracting structured and unstructured data insights on the engineering lifecycle platform at every desk and field user. Interviews with top leaders reveal that less than 20% of data in front of performers of tasks at the desk and computer screen is accurate. This discrepancy stems from insufficient standards or effective approaches for workforce education and skills development, inadequate standard technology platforms, and deficient standard information architectures. Addressing these issues is paramount for improving program delivery and decision-making. To counter this, a comprehensive, interconnected, and integrated approach is imperative for both new and existing programs, especially for AI-driven products (on a platform) and processes (supporting workflows).

This chapter will explore the challenges and solutions associated with implementing an AI ecosystem management and modeling capability across major programs and modernization projects. At the end of this chapter, specific solutions using available market tools, stressing the importance of proper guidance, teams, and approaches will be presented. Improving workforce capabilities at the desk level is crucial for adopting the AI ecosystem management and modeling approach. A mentorship program is essential to foster mastery of practice among technical staff.

A *Total Lifecycle* approach is an engineering lifecycle approach with a data fabric that will ensure "live" data access and work products based on truthful or "canonical" data. From there, AI approaches such as ML, DL, and generative AI can offer automation at scale, precise answers to questions across departments and programs, and ensuring for example, interoperability and sustainability, directly. The DoD and primes must harness Machine Learning (ML) for predictivity, enabling engineers and program staff to make informed decisions based on actionable insights. Deep Learning (DL), powered by neural networks, provides prescriptive answers to complex challenges, supporting practitioners with specific actions or options to achieve decision optimization. Generative AI can be tuned and/or trained to support each task in the lifecycle including analysis, generation, and evaluation of product or program information from data in the ecosystem. The result is a dramatic decrease in human error and the cost, risk, and schedule impacts of cascading unintended consequences experienced in almost every DoD program.

Regarding the prime value of an engineering program, the end result must not only deliver system performance but also user delight in the battlefield. AI models can boost these metrics with the involvement of warfighters and simulation and training experts early and often in the lifecycle. Once created, the AI-driven engineering lifecycle will require ongoing management to avoid drift and bias, as well as improvements and new AI models. For example, a program evaluator AI model could evaluate the entire engineering dataset to find conflicts, duplications, and perhaps a new critical path for the program manager and chief engineer. These AI management and development capabilities must be found on a full-featured AI development platform or "studio" that is on-premise and

can accomplish almost any task with on-board tools, while offering open APIs for the use of external tools, models, and data. Integration into the data center critical. AI model platforms can expedite AI model development using open-source or major vendor offerings as "accelerators" for discrete use cases in a program delivery. Examples include supply chain forecasting, prescriptive maintenance, and customer attrition prediction. These accelerators significantly lower the time and cost of development, training, and testing, thereby accelerating program delivery.

The AI platform will include a neural network for general development and a full suite for large language model (LLM) tuning and training. For some DoD business use cases, a private cloud solution will be sufficient for hosting an AI platform. However, almost every program will require an "on-prem" AI development and deployment platform. There are very few "on-premises" (on-prem) COTS offerings from AI vendors in the marketplace so the acquisition process will require very little time and effort. LSIs must immediately initiate the integration of an AI platform and data fabric. This leap at the LSI level should push other primes and subcontractors to do the same. The challenge will be with the DoD on the government side. This paper ends with a "prescription" for every LSI, most primes, DoD program, Program Executive Office (PEO), and Command, and formal Program. The value of the approach presented here has been proven to require investments in the right place with the right "boutique" expertise rather than a traditional rolled-up approach generating the classic large acquisition exercise.

8.2.1 Key Participants in Decision Support and Optimization

Interviews with top leaders generated a surprising result regarding their opinion on the accuracy of data in front of performers of tasks at the desk and computer screen of USG staff: less than 20% on average. If daily tasks, decisions, and deliverables are based on less than 20% accurate data, we can imagine that this is one of the root causes of the DoD challenge and the opportunity for relatively rapid improvement in program delivery. Figure 8.2 illustrates the wide range of roles that will need to participate in the new *Total Lifecycle* approach described here, using data and information in real-time.

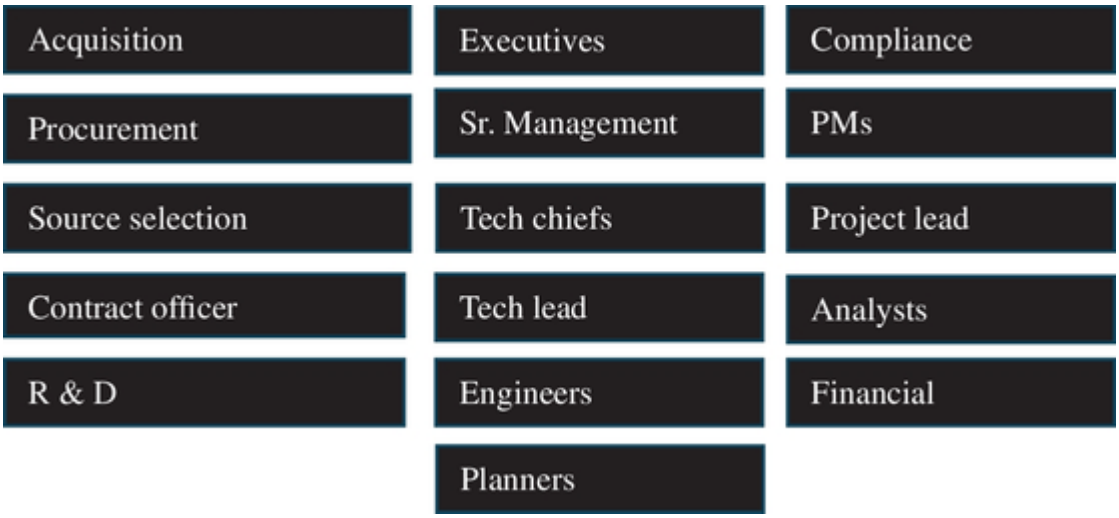


Figure 8.2 Decision support and optimization must be implemented at every desk, all the time.

8.2.2 Upgrading DoD Management and Operations to Meet the Needs of AI Injection

The appropriate on-prem AI platform can significantly enhance data governance and cybersecurity across the DoD's systems. Currently, the lack of data integration across tools and platforms creates vulnerabilities that can be exploited by adversaries. AI platforms offer continuous monitoring and vulnerability detection, ensuring real-time responsiveness to cyber threats. With a data fabric, the integration of AI and data governance frameworks for data access allows for more effective management and provision of data to the right system and program participants. This may be particularly relevant at the enterprise level for classified data, intellectual property, and patient information, for example. AI-driven data governance and cybersecurity can help protect sensitive information, maintain system integrity, and ensure compliance with relevant regulations and standards. By leveraging AI capabilities, the DoD can proactively identify and mitigate potential threats, thereby strengthening its overall cybersecurity posture.

8.2.3 AI Lifecycle Management

AI lifecycle management is crucial to ensuring that models remain functional, relevant, and secure throughout their use. AI platforms provide a foundation for managing AI models, from development to deployment, offering collaborative tools for model development, governance, and validation. A data fabric ensures that AI models receive the necessary data

flow from data lakes and external sources of information. It is essential to emphasize that the LLM is just the initial step in a potentially lengthy process of delivering a repeatable, reliable result or conversational session in a business or technical workflow. Transformation of the model via tuning or training is a significant effort requiring advanced tools and practical experience. Additionally, having the knowledge and tools to compare results across LLMs is crucial for satisfactory delivery. Another underappreciated aspect is the testing and acceptance of AI software. Testing AI software is vital to any AI work in the DoD. Unlike traditional software, AI results may not be repeatable, posing a significant challenge. "AI result repeatability" necessitates a specialized team and external support to establish a new testing regime.

AI-specific needs in the Total Lifecycle approach include:

- Continuous Learning and Retraining: AI models must be retrained regularly based on new data to maintain their accuracy and relevance in dynamic environments. Continuous learning frameworks enable this by automating data ingestion and retraining processes.
- AI Model Validation: Independent Verification and Validation (IV and V) are critical for ensuring that AI models meet the required performance standards. AI can be used to automate parts of the validation process, identifying discrepancies and areas requiring improvement.

The AI lifecycle management involves managing AI models from development to deployment, ensuring their functionality, relevance, and security. This process includes continuous learning and retraining, AI model validation, and addressing the unique challenges of testing and acceptance in AI software.

8.2.4 Solving the Classified Data Problem with a Data Fabric

Owners of rights of usage must also come together to contribute to a new layer of architecture that allows the workforce and systems access to data necessary for workflows and decision-making at every level. In our current disaggregated data and decision-making environment, most tools lack integration across lifecycle data creating significant challenges. The

AI ecosystem management and modeling approach includes a data fabric to address these issues by enabling Program Managers to effectively manage classified and unclassified data within large programs. While this paper does not delve into the specifics, future white papers will explore methodologies to achieve this goal comprehensively.

8.2.5 Sample Application: AI Autonomy

Autonomous systems, powered by AI, will become increasingly vital in the future of defense. AI-driven autonomy can be applied to unmanned systems, enabling them to execute complex missions with minimal human intervention. AI models trained on extensive datasets can anticipate mission needs, predict system failures, and optimize resource allocation in real time. For example,

- Autonomous Mission Planning: AI can automate mission planning by processing vast amounts of sensor and environmental data, determining optimal strategies with minimal human input. These models can adapt in real time, enhancing mission success rates under dynamic conditions.
- Prescriptive Maintenance and Logistics: AI models can predict equipment failures before they occur and prescribe the time, data, spare part, and technician based on a wide range of variables, ultimately delivered by a neural network, thus solving the entire problem without a man-in-the-loop.

The current disaggregated data and decision-making environment presents significant challenges due to the lack of integration among numerous tools. The AI ecosystem management and modeling approach includes a data fabric to address these issues, enabling program managers to effectively manage classified and unclassified data within large programs. While this paper does not delve into the specifics, future white papers will explore methodologies to achieve this goal comprehensively.

8.2.6 AI and Change Management

One of the most significant barriers to rapid AI adoption is change management. In defense programs, where hierarchical structures are deeply ingrained, AI-driven systems often encounter resistance unless the change is meticulously managed. An AI-enhanced change management process must be established to facilitate rapid feedback loops and guarantee real-time updates across all system components. By integrating AI into change management workflows, the defense ecosystem can optimize communication, minimize delays, and mitigate risks associated with implementing new technologies. This approach will enable the defense sector to more effectively adapt to the transformative potential of AI, ultimately enhancing operational efficiency and competitiveness.

8.3 How to Implement the AI-driven Ecosystems Management and Modeling Regime

8.3.1 PPTI Framework Has Assisted Executives for Decades

The problem can be articulated through the lens of PPTI, slightly adjusted. The challenges are:

- People: Insufficient standards or effective approaches for workforce education and skills development.
- Procedures, not Process: Lack of standardized and automated methods, procedures, and techniques for tasks.
- Technology: Inadequate standard technology platforms, tools, or transformation capabilities.
- Information: Deficiency in information architectures, knowledge management, and AI governance.

8.3.2 Solutions Based on Insights from the Field

The DoD, contractors, and commercial vendors are struggling to maintain US competitiveness due to rising adversaries' ability to deliver faster, better, cheaper, and in some cases, smarter (e.g., AI and cybersecurity).

Through decades of interviews with senior executives, warfighters, and leadership at LSIs and primes, we have identified three areas of need:

- Digital Engineering for the Lifecycle: This regime should emphasize the collection, storage, and linkages to lifecycle data including rational and architecture elements, which will lead to flexibility, agility, and rapid iteration to keep pace with adversaries.
- Rigorous Change Management Regime: A robust and expedited change management regime is essential to accommodate the dynamic nature of modern warfare and technological innovations. This regime should facilitate seamless integration of new technologies and methodologies, minimizing delays and risks.
- Use of Architecture at every Desk and Field Operator: Incorporating architecture into every task and decision-making process is crucial for ensuring interoperability, efficiency, and effectiveness across the defense ecosystem. This approach will enable better collaboration, resource allocation, and mission success.

Other feedback from current and former senior officials, staff, and industry players describe specific areas of improvement:

- Innovation Deployment: Rapid deployment of innovation is nearly impossible.
- Rapid Response: Instant impact analysis and change at the desk and with the responsible teams and approvers.
- Concurrency Issues: Significant lag between ever-changing program platforms and training systems and external, currently interoperable services partners,
- Portfolio Management: Collaborative management of cross-program dependencies from executives through programs out to LSIs and vendors.
- Data Accuracy: Decisions are frequently based on low accuracy at the government desk, often less than 20% accurate data; decisions at the desk can thus generate huge cascading errors in the program.
- Acquisition Speed: The acquisition process is slow and manual with an overworked workforce; an information-driven, online, system with AI models helps evaluators to generate the right solicitations for new ac-

quisitions or review the existing contracts and program data for modifications.

- Requirements Justification: Requirements Rationale cannot be one statement, per past standard procedure. It must include use cases, stories, requirements from previous projects, wireframes, and loose data that drive a future requirement into the program. Once such a collection is created, it is important to save these justifications to generate other requirements or change or waive requirements, design, or program elements.
- Legacy Systems Insights: We can not acquire valuable insights from existing legacy systems, databases, and data lakes.
- Interoperability: Interface specifications alone are insufficient; a fully executed integrated UAFML model is necessary to identify the interoperability necessary for a task to be performed (and backup procedures thereof). Not everyone in a program needs to be an expert architect which is a systems engineering effort.
- Communities of Interest: Engineers cannot cooperate across disciplines for design tasks. They must be able to published changes to identify any errors or risks. From there, the leader of the discipline or "blog" may inform a program manager of the errors and risks to help avoid unintended consequences.
- Requirement Sharing: Requirements are rarely linked to specifications and related architecture, specifications, rationale, and test cases, and made available to the larger community. Derived requirements and specifications should be available live to program participants.
- Change Management: Current change management processes are slow and imprecise, causing delays. Many in the DoD have expressed the urgent need to form rigorous change management without the effort and delays of manual approaches. Some organizations have enjoyed 10–20× speed in technical and programmatic change cycles with the Total Lifecycle approach.
- MBSE Adoption: Most of those interviewed identified adoption challenges for MBSE and willingness of leadership to view, understand, and use the diagrams for decision-making. Rapid solution with an AI platform and data fabric.
- Supply Chain Optimization: Automation and predictive capabilities are needed to reduce costs in every workflow in the business and pro-

gram and the value chain and kill chain/web in the battlefield.

- End-User Collaboration: Many DoD systems are delivered with users being disappointed with products offered. Early and continuous business or warfighter user is missing in most Programs. Users should be able to participate at the project and throughout the lifecycle.
- Workforce Development: Effective workforce development and upskilling require engaging methods. The toolset and methods described here will require a proven adoption plan with continuous training.
- Attracting Talent: Graduate students need more domain knowledge, practical skills, and tool experience to contribute effectively. The DoD must invest in offering undergrad students and their professors access to Total Lifecycle tools, methods, and training.
- Prescriptive AI for Supply Chain: Prescriptive AI is necessary for optimizing the supply chain, value chain, and kill web/chain efficiently. These decision optimization prescriptions are the “holy grail” of AI whereby a specific action can be taken having considered all the services and elements necessary for that action at the timeframe in question. The AI system can prescribe options to a commander, for example, considering the cost, effectiveness, and logistics and the many other factors of a mission operation in the battlefield.
- Collaborative Digital Engineering and Thread: The Total Lifecycle is instantiated by the AI-driven engineering ecosystem management solution with asynchronous capabilities for live data access in a collaborative remote lifecycle platform allowing global interaction, for example, across geography and time zones.
- Sustainment Planning: Early and optimized planning for system sustainment is essential to manage costs effectively. There will be significant impacts to requirements, design, and testing as “design for sustainment” will be critical in future programs. Sustainment experts are required to participate prior to the program start.
- Generative AI: The advent of Large and Small Language models has made many in the market an AI “expert” without an understanding of the transformation and testing process required. It is certainly not a DIY effort. Partnering with the right delivery firm will be a boon to business, training, and weapons systems programs and solutions, some of which are described in this chapter.

8.4 Key Elements of AI-driven Ecosystem Management and Modeling

A Total Lifecycle approach, instantiated by an AI-driven ecosystem management and modeling solution, to supported lifecycle processes, not just delivery of AI software products. This approach aims to achieve significant speed, precision, cost efficiency, interoperability, concurrency, DoD-LSI communications, workforce attraction, retention, and upskilling, and enhance business or warfighter user delight. The AI-driven ecosystem management and modeling solution will address the challenges identified in the previous sections, providing a comprehensive framework for managing defense programs and systems. This approach will leverage AI capabilities to optimize decision-making, streamline workflows, and improve overall performance across the defense ecosystem. By adopting this Total Lifecycle approach, the DoD can better position itself to maintain competitiveness and ensure mission success in the face of evolving threats and technological advancements.

The “digital thread” in this context refers to an organized set of links and associations for the live presentation of information to the user, enabling traceability during changes. These links include attributes of model elements in architecture and program elements, such as use cases, stories, requirements, test cases, military specifications, and compliance guidelines. An AI-driven ecosystem management and modeling solution can achieve a live, digital engineering environment in a program, portfolio, enterprise, or LSI contractor and vendor tiers by prioritizing the following capabilities:

- Real-time data access and integration: Ensure seamless data flow across tools and platforms, enabling effective decision-making and collaboration.
- AI-driven decision support: Leverage AI capabilities to provide actionable insights, predictive analytics, and prescriptive actions for improved decision-making.
- Continuous learning and retraining: Implement continuous learning frameworks to automate data ingestion and retraining processes, maintaining model accuracy and relevance.

- AI model validation: Utilize AI to automate parts of the validation process, identifying discrepancies, and ensuring performance standards are met.
- Change management: Establish an AI-enhanced change management process for rapid feedback loops and real-time updates across system components.
- Architecture-driven approach: Incorporate architecture into every task and decision-making process to ensure interoperability, efficiency, and effectiveness.

By adopting this AI-driven ecosystem management and modeling solution, the DoD can create a permanent improvement in program and portfolio management, leading to significant leaps in joint enterprise warfighting capability and that of allied partners. This approach will ensure that decisions are optimized at every desk within the ecosystem, maintaining US superiority (Figure 8.3).

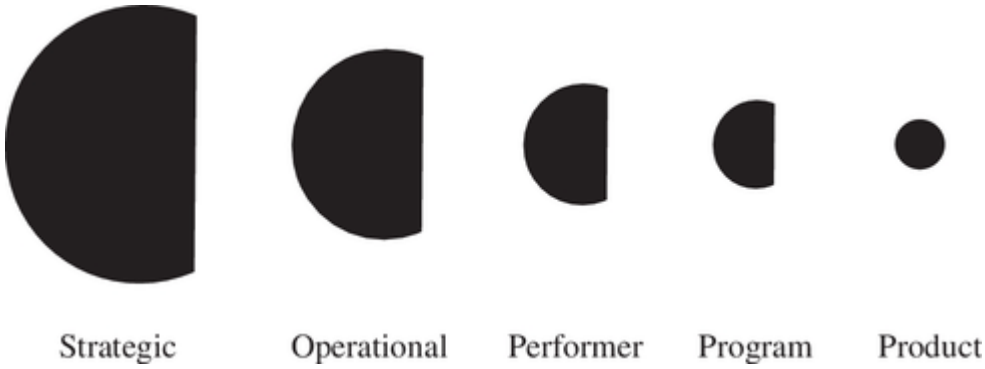


Figure 8.3 Value of AI at levels in the ecosystem must be considered.

8.4.1 We Must Integrate Program Management and Systems Engineering

Program Management and Systems Engineering must not continue as separate disciplines. Decision-making must improve through an integrated perspective. Today's programs require intense internal and external integration, demanding that Program Managers exercise "Program Engineering Management" abbreviated here as the Total Lifecycle management. This term describes the full integration of program management, systems of systems engineering, product design, acquisition, and the delivery of capabilities serving missions and objectives targeted.

Decision-making must improve through an integrated perspective. Today's programs demand intense internal and external integration, necessitating that Program Managers exercise "Program Engineering Management," abbreviated here as Total Lifecycle management. This term encompasses the full integration of program management, systems of systems engineering, product design, acquisition, and the delivery of capabilities serving missions and objectives. The ecosystem concept, illustrated in Figure 8.4, enables the workforce to comprehend technical impacts and design consequences, thereby enhancing decision quality at every desk. A recommended resource for further understanding is "Integrating Program Management and Systems Engineering," edited by Eric Rebentisch (Wiley 2017). By adopting this integrated approach, the DoD can better manage complex programs, improve decision-making, and ensure the successful delivery of capabilities serving missions and objectives.

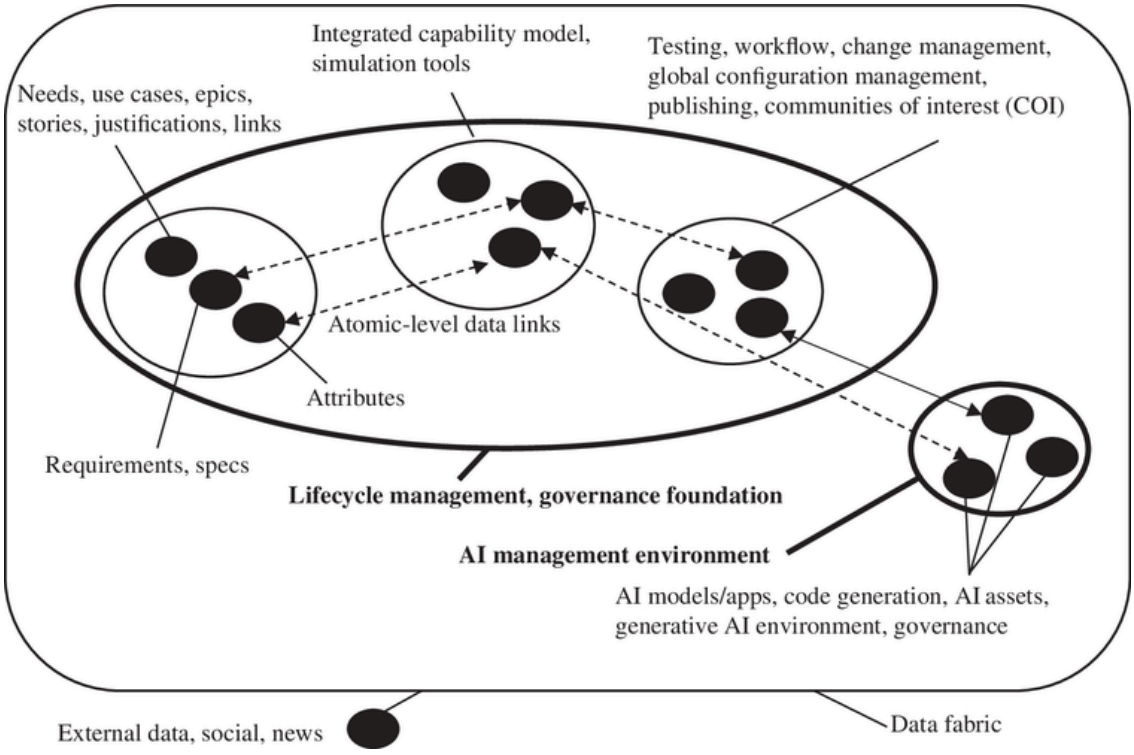


Figure 8.4 The program lifecycle must function like an organism within a larger ecosystem.

8.4.2 Harmonize Agile Development with Traditional Acquisition

Agile Development: Software development with agile enables sponsors to participate in progressive deliveries that occur in days and weeks vs. months and years. Agile techniques can be included in a large program or

project systems engineering "Vee" (phased project delivery). Projects must "negotiate" the mix of agile and project approaches prior to program start to ensure success. Figure 8.5 illustrates the basic efforts required to plan and execute an agreement on methods and subsequent participation prior to the start of a program or increment.

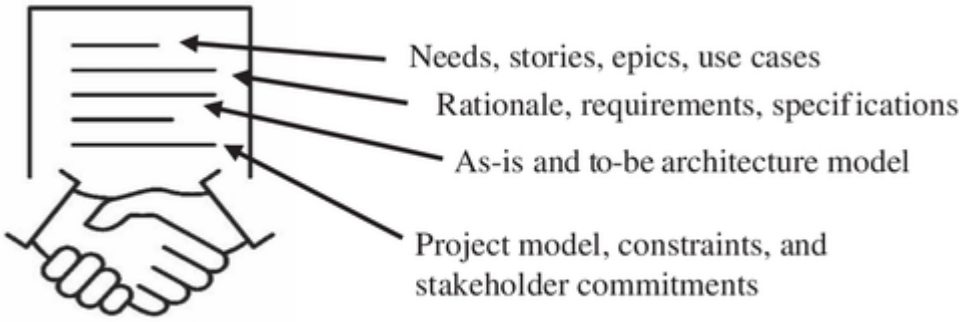


Figure 8.5 An upfront contract is necessary for agile and waterfall approaches to coexist.

8.4.3 Ensure Requirements Quality for Testable Language

A majority of interviewees believe that requirements are poorly written, complicating testing and LSI responses. The same issue is prevalent among LSIs when selecting first and second-tier suppliers/subcontractors. Generative AI models can help enhance the requirements of language and eliminate duplications, adhering to INCOSE and other standard principles and guidelines. By preserving the original inbound requirement statements and linking them to the improved requirements language, the live platform enables sponsors to map their requests to delivery more easily. This approach ensures better communication, reduces misunderstandings, and streamlines the collaboration between stakeholders. Incorporating generative AI models into the requirements management process can significantly improve the quality and consistency of requirements, ultimately leading to more efficient and effective program execution.

8.4.4 Increase Concurrency in Simulation and Training

As platforms, weapons systems, and instruments evolve over the life of a development program, live virtual constructive training and synthetic environments must keep up with some or all those new features to maintain

concurrency for the purposes of training. Similarly, variance management must be managed in a rigorous manner. Concurrency can be achieved with a platform to track, analyze, and manage changes and eventually full configurations of each variant leading to more rigorous mapping (and testing) of a program's platform training requirements. Figure 8.6 identifies standard simulation and training elements that will be affected by concurrency issues.

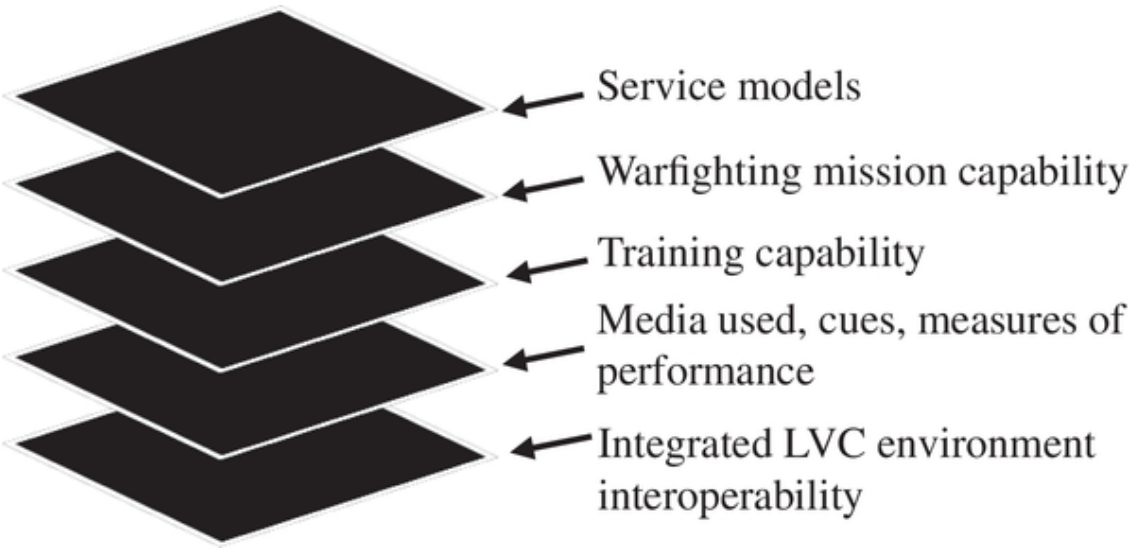


Figure 8.6 Continuous concurrency can be achieved in simulation and training programs.

8.4.5 Enhance Cybersecurity with MBSE

Rising cybersecurity threats, technology, and subsequent regulations have slowed delivery progress in many parts of a program. Using architecture models in UAFML, detailed attributes in data exchanges and information systems, risks and exposures in DoD systems can be identified for decision support, conservation of security against changes in the program, and innovation opportunities to preserve and combat these challenges to optimize system safety and maximize reduction of hazard to troops. In 2022, the US Marine Corps highlighted the need to use systems engineering and MBSE to address these cybersecurity challenges.

8.4.6 Embed Human Factors into MBSE

Modeling human factors in architecture models is a growing capability that can enable the expression of human factor attributes for planning, design, and change management, integrating them into every key engi-

neering discipline to ensure Total Lifecycle delivery of a program capability into the battlefield and other operational environments.

8.4.7 Implement Early End-user Participation

Programs must involve end-users early on, allowing for higher acceptance of deliveries. A live data, asynchronous platform with remote access can enable more diverse and impactful end-user participation from the field.

8.4.8 Prioritize Sustainment Planning from Program Start

Sustainment must be a core focus from the inception of programs. Sustainment experts should participate in driving requirements and design throughout the program lifecycle and be able to utilize the Total Lifecycle approach and solution.

8.4.9 Understand and Utilize “V and V” and “IV and V”

Verification (requirements met) and validation (value achieved) must be formally addressed and utilized throughout the program lifecycle. Agencies like NASA use a significant amount of the program budget with contractors for Independent Verification and Validation (IV and V), ensuring high-precision system delivery.

One of the most critical elements in AI development to support an engineering lifecycle task or other AI application development, requires new forms of testing must have its own set of “AI quality” staff, methods, and techniques, similar to the V&V describe above, and with new tools other AI models in order to accept the software into production.

8.4.10 Enable Impact Analysis for All

One of the leading daily use cases for a Total Lifecycle approach is impact analysis at the desk. Very few organizations have succeeded in implementing architecture as a daily tool or “cockpit” to simply understand the “as is” or “to be,” target architectures. Once inspected, the program partic-

ipant at EVERY level must be able to perform impact analysis on a design task, upgrade, or change. Imagine not understanding the impacts of a fix to a test failure and the dependencies impacted across the program or ecosystem. Selection of tools that allow this to occur is critical or there cannot be a Total Lifecycle operation.

Tools must allow every program participant to perform impact analysis quickly and easily for design, development, change management, and test failure planning, as key examples. Atomic-level links offer traceability across the UAFML architecture and over use cases, requirements, test cases, workflow tasks, and integrated master schedule (IMS) events. There should be higher-level links to cost, risk, and compliance documents and artifacts as mentioned earlier. The right tool vendors use open services for lifecycle collaboration (OSLC) technology offers linked data functionality.

With the right vendor solution and additional guidance from systems engineering experts, impact analysis can be further enabled by using methods to allow various levels of users to utilize the architecture model. For example, an upgrade to a processing chip in the weapon system to process data more quickly could result in a buffer overrun on the aircraft computer (a real example). Clearly, the computer engineer will want to view the architecture and make changes (as he will be educated in architecture and UAFML modeling). This engineer will identify the bandwidth specifications of the aircraft computer system. In the approval process, a quality engineer may right-click on the architecture to perform impact analysis, review the concept, and further generate a test plan. This engineer will also have been educated on architecture and UAFML modeling. A deputy program manager may download a report containing diagrams and spreadsheet data to approve the proposal or upload a spreadsheet of a new project, with cost, risk, and schedule impacts that will change the architecture and generate an approval cycle. Similarly, a contracts officer may identify impacted contracts by inspection or spreadsheet downloads. Over time, the ideal Total Lifecycle operation will have most of the workforce able to use the model for analysis with or with most having very little architecture and UAFML modeling expertise, but rather the ability to use the model for most tasks, with the power of impact analysis. Figure

8.7 illustrates a high-level view of what should be architected into formal UAFML models. In this graphic, a program manager can quickly manage what is inside his program while ensuring interoperability for concurrency with external services as they change over time.

Figure 8.7 Simple illustration of elements and associations that can be used in a UML tool for impact analysis.

Many architectures are missing the services model views that identify how system elements collaborate to deliver, or accept, a service. A service may be a satellite product necessary to perform a mission (which would be outside the program and enterprise but within the ecosystem). Service models also provide service-level agreements to identify the attributes of the service being performed. For a close air support mission, for example, ground troops may want to understand the probability of the service being provided, response times, and size of the blast.

Service modeling also brings in senior-level management into the program to help them visualize the value being promised by the program and the larger enterprise capability delivery. It will also assist program managers, contractors, and executives in negotiating changes to ensure that the service and SLA are not impacted (or agree to the level of change in the service or SLA). Modeling a web of elements that make up a service to support a mission, such as close air support in a battle area, can enable also high-level stakeholder understanding and support more rigor to table-top exercises or battlefield simulations. Service model diagrams draw executive and management participation and interest into viewing architectural models, which is an enormous challenge today.

8.4.11 Conduct Rigorous Prototyping Projects with Top Experts

Prototyping must be managed with more rigor and expertise to drive the value of innovation into programs and eventually the battlefield. Although some prototypes are experimental exercises, the vast number of prototypes should switch to being testable pilots vs. throw-away experimental systems and software. An AI-driven ecosystem management and

modeling solution can manage prototypes and pilots with rigor similar to formal program delivery.

8.4.12 Engage Small Businesses in Ideation Diversity

Generating innovation within an operational workforce and vendor community is already challenging. Ensuring diverse thinking avoids groupthink and drives rigor and innovation. Effective ideation processes are needed to filter and evaluate innovations, determining their potential as development projects. These innovative efforts across a community of diverse thinkers and small businesses can also be managed and integrated into programs with less friction. The solution should also support eliminating the valley of death from basic research and applied research results and program-ready prototypes and pilots.

8.4.13 Implement Virtual and Augmented Reality (VR/AR) and Metaverse Tools in Program Engineering

As model and data complexity grows, VR/AR tools can enhance a user's interaction with the digital engineering environment, aiding in analysis and communication with greater speed and precision. Investment in these technologies can expand understanding and adoption of lifecycle capabilities and drive additional rigor and user delight at delivery. AI will be a foundation for these efforts and a government-owned or LSI-owned AI development environment will be key to containing and managing AI software to meet compliance.

8.4.14 Accelerate Source Selection with Remote, Model-based Capabilities

Modernizing source selection and evaluation with new, centralized, integrated engineering lifecycle platforms can speed up the evaluation and award bandwidth, potentially 2–3x and lower the number of protests.

8.4.15 Use Integrated, Model-based Compliance

Demonstrating compliance with expanding laws, regulations, policies, and guidelines is increasingly difficult. An integrated compliance capability within a lifecycle platform can streamline this process, ensuring rapid and successful audit outcomes. A combination of linking federated data and the use of generative AI models will offer "live" compliance checks at the desk and in the field for internal inspection and audits that may occur. An elastic search on large repositories can now be used for structured responses to queries as well as conversational AI. These AI Digital Libraries will be a key application offering rapid time-to-value to Program Managers.

8.4.16 Fully Integrate AI Management into the Total Lifecycle

As we have discussed, AI can provide valuable insights from data across the lifecycle. An on-prem AI governance environment is required to contain, manage, and operate AI applications securely, knowing that open APIs exist to access other AI sources, but under governance from the central AI platform.

8.4.17 NLP and Pretrained COTS Models

Off-the-shelf NLP, vision-detection, and other pre-trained models or "cartridge" add-ons can quickly offer analytical capabilities without further turning and training. Some of these COTS models can understand 13 languages and identify emotion and sarcasm in social or internal office communications, for example, to gain business insights from data analysis.

8.4.18 Machine Learning

ML offers automation at scale, providing high-speed, precise answers to questions across departments and programs. Since ML automates the analysis of large amounts of historical data, it also delivers predictivity as a decision-support tool.

8.4.19 Deep Learning (DL) and Neural Networks

DL provides prescriptive answers to complex challenges, supporting engineers and program staff with specific actions or options. Using a neural network, traditional DL can: (i) generate a prescriptive approach to a task or decision (decision optimization), and (ii) "create" a new idea, concept, or design that no human could have developed. We have seen the former being utilized for prescriptive maintenance of aircraft while the latter has been used to design a new form of highly efficient rocket engine.

8.4.20 AI Model Accelerators

Accelerators are pre-created foundation models that can speed AI model development. Examples include supply chain forecasting, prescriptive maintenance, and customer attrition prediction.

8.4.21 Generative AI Concepts

Much has been published on generative AI and LLM but the opportunity for the DoD will be missed if not properly managed using the guiding principles of systems engineering and tools thereof, as described in this chapter. Excessive media attention has focused on the latest LLM development. The LLM is just the start. Almost any LLM can be initially used as an initial step of a workflow that includes the task of experimentation, iteration, tuning, and in some cases, training the model and how the user will use RAG to generate results for the task in question. Some efforts will require the creation of a small language model. Others will require on-prem agents to be created. Further, some use cases will require a web of AI models and agents to deliver a result. Details on how to deliver generative AI will be discussed later in this chapter.

8.4.22 Implement Digital Twins with Digital Threads

Another way to view the engineering lifecycle methods presented here is that it is a digital twin of the program. The links described above are the "digital threads." Digital twins of physical or operational systems using

parametric “twinning” is a different and useful endeavor, assuming that the ROI is justified.

8.4.23 Leverage Quantum Computing

Quantum technology can solve practical problems beyond the capabilities of classical supercomputers. Integrating quantum computing into the AI ecosystem management and modeling solution will enhance data analysis and problem-solving. One can utilize the engineering lifecycle methods in this paper to identify whether the cost of quantum project experimentation can be justified.

8.4.24 Implement a Data Fabric

Data fabric architecture integrates various data pipelines and cloud environments, facilitating large-scale data analysis. This foundation is essential for achieving speed, accuracy, and data security. For both the engineering lifecycle and AI development and operations, a data fabric will ensure that the right data is presented to the right human/system in everyday project workflows, program reviews, and decision-making.

8.5 Enhance Workforce Development and Mentorship

Improving workforce capabilities at the desk level is crucial for adopting the AI ecosystem management and modeling approach. One element of an engineering lifecycle is to drive technical staff toward AI proficiency. A mentorship program is essential. To acquire new hires, a new generation of workers can be attracted to an AI-based work environment and the opportunity to gain certifications.

The challenge of acquiring the most qualified engineers vs. those that simply enter with operational experience in the field is leading to gaps in program efficiency. In Figure 8.8, we identified a cake analogy where each layer is critical to the initial value that a new hire brings to the organization. Undergraduate engineering students should use this as a guide

to compete for the job opportunity they desire at one level higher than the typical graduate.

Figure 8.8 A “Cake” metaphor for mastery in engineering capability in the workforce.

8.5.1 Layer 1: Domain Knowledge

Understanding the defense industry, and the organization’s vision, mission, strategy, and culture is fundamental.

8.5.2 Layer 2: Practice Knowledge

Proficiency in standard methods for solving problems in a discipline such as electrical engineering, lethality, accounting, or materials science.

8.5.3 Layer 3: Tools and Platforms

Proficiency with vendor solutions that support the capabilities discussed in this paper (most tools do not have the ability to be operated in the manner described in this paper, while those standardized by the large primes fill the requirement).

8.5.4 Layer 4: Techniques

“Techniques” are the most critical element. Techniques are prescriptive methods learned or developed by experience or mentorship to perform tasks or tackle complex challenges that come with a new role.

8.6 When Can We Acquire Dramatic Speed and Precision?

We often hear, “oh, that is a dream state” or “that’s early to need” and “we already do all this.” Fortunately, the DoD can get this right, and right now with a rigorous engineering approach supported by the latest, modern approaches, today. If adversaries continue to advance with state-sponsored, single-mindedness, and authoritarian program controls, waiting just half a decade could irreparably disable the US advantage. A starting

point for rapid implementation can be identified by our principles, each of which has been described above:

- Live Data and Information for every task and decision: Complete set of integrated lifecycle capabilities with the proper integration points across the ecosystem, with linked data at the atomic level across relevant ecosystem elements and model-based decision support.
- Secure and Compliant AI "fortress:" AI platform and development/operational environment, on-prem and strictly governed.
- AI lifecycle and models for use in business, training, platforms, and weapons.
- Data Access and Permissions: A data fabric foundation to allow the right data to be available at the right time for the right workflow to the right person.

8.7 Which Elements Exist in "AI Ecosystem Management and Modeling?"

To reiterate, the benefit of an AI-Driven Ecosystem Management and Modeling approach lies in optimizing decision optimization from top to bottom, and across primes, subs, suppliers, and partners. In the context of Dr. W. Edward Deming's perspectives, *systems* in an organization are how you operate and require the acquisition of expertise for the design and operation of the new capabilities. Discrete *Solutions* that support these systems are presented at the end of this paper as high-level architecture and commercial tools.

8.7.1 Integrated Object-oriented Model(s)

An "MBSE" model must be:

- the "as is" canonical model of truth accessible and usable by the workforce/team members,
- the "to be" as a means to simulate future possibilities for planning, analysis, and decision support, and
- defines dependencies and capability levels with data linked to associated programs, services, and operations as well as external data

sources

MBSE models are created as an integrated capability model generated by a profile and language standard to their Domain. The DoDAF 2.x framework defines the standard model package and taxonomy, and the profile standard is UAFML. As stated earlier, the key to success is using special *techniques* that lead to success in program engineering, not just model creation. Lack of the "SE" in MBSE is a root cause for failure in modeling efforts. "We never open the modeling files after they are approved and delivered," say many program managers. New forms of data presentation are now possible to persuade managers to view architectures, by exporting diagrams to mind map visualization tools, for example.

The right models can support program engineering continuously with the analysis of "as-is" and "to-be" models created for planning, analysis, acquisition, design, build, V and V, and user acceptance. Not everyone needs to be an architect. *"Non-modelers" should be able to use the right models with the right data and the right views to conduct impact analysis (without being an architect) quickly. These techniques can be acquired with knowledgeable partners in industry.*

8.7.2 Integrated Modeling with the Right Profile, UAFML

The UAFML profile has been implemented in a few object-oriented modeling tools. It is the standard created by the DoD, the UK Ministry of Defense, NATO Allies, and other allied countries as well as the object management group among others in industry and academia, working groups, and independent advisors. The use of this profile is absolutely necessary for AI ecosystem management and modeling, i.e. not UML or SYSML alone.

To ensure readability, the organization should embellish the UAFML Profile with a new standard set of graphical decorations on most stereotypes as opposed to the hard-to-read "boxes and arrows" in the implemented profiles, which lead to even lower adoption than experienced today. An example of success in this regard was demonstrated in the early

days of DoD profile efforts for a UPDM standard by IBM in the form of their UPIA profile.

As mentioned earlier, a critical capability is atomic-level traceability between a UAFML model element and other program element, such as requirements, justifications, tasks, test plans, internal artifacts, external assets, and external applications, which we call fine-grained traceability. Traceability enables impact analysis at every decision and work product, which may be the most critical aspect of AI ecosystem management and modeling capability. Figure 8.9 illustrates how graphical icons in architecture can make it more digestible for decision-makers.



Figure 8.9 Example of stereotype decorations as opposed to a myriad of “Boxes and Arrows.”

This *Design Management* concept brings to engineers a bidirectional association and a “live” presentation of data and model views. This atomic linking across architecture, requirements, justifications, and test plans is the largest factor contributing to speed to field and precision. Unfortunately, few tools offer this capability. In AI ecosystem management and modeling management, you can associate all software and systems designs with other lifecycle resources such as requirements, change requests, and test artifacts. Software architects, system engineers, devel-

opers, and deployment planners can collaborate with peers, subject matter experts, clients, and other stakeholders over the internet.

When integrating UAFML models with program elements, one can set up links between design elements, other designs, and lifecycle artifacts (architecture management, requirements management, change management, and quality management.) The bidirectional traceability between requirements and designs enables you to easily see and understand the impact of changes to the requirements or designs. You can also create and view impact analysis diagrams to examine the links to and from specific target elements in a design. One benefit resulting from this traceability is a rigorous form of Change Management.

Live reporting and document generation are also important. There should be embedded document generation features and templates to generate and share design documents, specifications, and reports. A web client can be used to generate reports, whereby project members and stakeholders can easily access information in the models and projects.

Live, online graphical modeling features are beneficial for creating sketches of architecture, and design ideas, wireframing, use case analysis, and descriptions of stories.

“Sketches” and “wireframes” can describe functions, tools, and data to drive an understanding of the rationale and justification of new or borrowed requirements and assist in “pre-design” assessments and evaluations of the approval process of requirements thus supporting the requirements generation, requirements change management, and design change management tasks as well as test case development and issue resolution and deconfliction.

8.7.3 Integrated Engineering Lifecycle Platform Foundation

A modern collaborative governance platform for program, engineering, and portfolio management is required to offer security, governance, and integration. The best collaboration platform can host a standard set of tools that ensures security, roles, shared services, and all the expected capabilities of a modern platform (including data management, link man-

agement, reporting, and open; each tool allows linkage at the atomic level and allowing other tools to integrate live and at the atomic level via OSLC where possible.

8.7.4 Integrated Justifications That Drive Requirements Generation and Traceability

"Justifications" are "rationale" data that describe existing or new requirements and provide valuable information to users by offering a linked set of collections of rich data and links to other assets, views, data, and even applications. Without Justification, impact analysis cannot be fully executed, and thus wrong decisions can be made, and unintended consequences are experienced.

Consider a functional requirement that must be changed due to a test failure and the determination to relax the requirement. Using linked Collections of rationale data, the engineering and program management team can inspect the justification, or rationale "collection" which linked to a formal requirement to make a decision. These collections may be linked ot documents and drawings, specifications, LRPGS, and other program or ecosystem data.

8.7.5 Integrated Program Method for Standardization, Improvement, and Business Continuity

How do we stand up a new program today? Cut and paste from a previous program? Or attempt to reinvent from scratch.? Neither approach is efficient or timesaving. The a new approach is to set up the program in method composition tool and learn and continuously improve the method as new program as launched.

A *Program Method* includes processes, work breakdown, workflow, roles, and tasks in advance and automatically injects the right tool into the workflow management capability. In other words, the program method can be saved to automatically generate a baseline program workflow and IMS for analysis, negotiation, and initial submission.

A process, in this context, is the collection of roles, practices, rules, permissions, and guidelines that you use to organize and control the workflow for a project. Robustly implementing a process, live and asynchronously, alters the delivery speed and precision, making conformance with the process more reliable.

The benefit is a rapid stand-up of the program, ensuring improvement for future programs, and achieving business continuity holding vital information as the workforce turns over.

8.7.6 Integrated Requirements Generation, Traceability, Management

A proper requirements regime enables you to capture, trace, analyze and manage changes to requirements while maintaining compliance with regulations and standards, thus avoiding costs and time of unintended consequences discovered later in the program. Unlike old requirements tools which are monolithic and not open to external use and management, a platform-based requirements capability offers the following benefits:

- Ensure transparency, minimize rework and increase reusability
- Ensure traceability and visibility of changes
- Ensure productivity by using version and variant management
- Strengthen collaboration among multidisciplinary teams

As engineering managers, engineers, and acquisition leads, the DoD must have requirements linked to justification collections of rationale data, fine-grained linking to model elements, and links to LRPGs and compliance data. We must also have automation for requirements generation to guide and track the approval process. We can develop epics, stories, wireframes and use cases and record them here linking them to requirements for understanding for design and change management processes. We can thus optimize communication and collaboration among teams and stakeholders to maximize productivity and quality.

Using an expanded requirements capability, we must be able to reuse valuable engineering work by effectively managing product versions and

variants. We must be able to effectively manage product variants for block delivery and foreign military sales.

8.7.7 Integrated Requirements Quality Scanner

Half of all engineering errors are due to poorly written requirements, and every industry knows that costs and delays rise exponentially when errors are found later in the program. Clear requirements also drive quality and test planning ensure better communication across thousands of program participants.

Today, it is possible to “clean up” requirements and derived requirements and spec language language while maintaining the original. One may use an AI NLP model to improve your requirements language, making them “testable” and traceable based on INCOSE guidelines. Each Program will be able to save time, optimize resources when developing requirements, and ensure verification testing, validation, and acceptance by users and customers.

- Reducing the cost of defects
- Reducing the cost of manual reviews
- Using the INCOSE Guide for writing requirements to help train junior engineers
- Speeding time to market by catching requirements errors early in the engineering process
- Clarify the services being delivered to stakeholders

With precision requirements, a program can also manage changes and impacts with rapid response to innovation injection, changing services, new compliance regulations, and other changes that occur during a long-term program effort.

8.7.8 Integrated Portfolio/Project Management and Workflow Management

Project portfolio management (PPM) is a crucial capability that enables a management hierarchy from the organization’s top down to the task level. The workflow can be imported from the Program Method design capa-

bility to initiate the Integrated Master Schedule (IMS). PMs, Chief Engineers, managers, and acquisition staff utilize workflow management to track their work, share changes, and collaborate asynchronously. Depending on access and roles, team members can provide input by iteratively replanning tasks and projects in the IMS to determine the most optimized approach forward. Communities of Interest may be set up for specific engineering disciplines, management levels, or project.

A modern workflow management regime allows components to be process-aware, enabling the addition of rules for behavior in the form of preconditions and follow-up actions. The processes must be highly customizable, such as the approval process for a requirement change. Configurable processes can enforce different rules at various points in the program, for example, employing a more restrictive process toward the end of a program to minimize the risk of introducing regressions. By implementing a robust PPM and workflow management system, the DoD can enhance collaboration, improve decision-making, and ensure the successful execution of complex programs and projects.

8.7.9 Integrated Change Management

Change management is the process of managing and controlling changes to a product, system, or project. Version control is managing and tracking changes to software, documents, or artifacts. Configuration management is the managing and tracking of the configuration and settings of a system, application, or infrastructure. Each concept serves a unique purpose and is used in different contexts, but they are all essential components of a successful software development and project management process. Change is often considered the primary root cause of DoD program delays, even more than a challenged requirements management regime. To address this, every leader and manager from the government to primes and subcontractors should conduct an initial analysis at their desk first when given an SMR, defect, or other change in a live program. This responsibility is critical to speeding the program schedule.

At the desk, a manager can immediately determine where in the program the defect exists, if it exists in their program, the impacts on the program,

service, or technical configuration at a high level, and to whom the defect or issue should be assigned. This ten-minute initial effort can cut the time of a change process from the very beginning by days. With atomic-level traceability, the Chief Engineer can use the latest technology to communicate the defect's relevancy to the right participants with near 100% accuracy. The assignment can be properly sent to the right person, and the system should be able to handle an "*n*" number of levels of such assignments, diving down to IPT leads, LSIs, and suppliers. PMs have experienced 20× speed and 3× accuracy of data on the desktop and in the field (to near 99%) with these techniques and systems in place. By implementing robust change management methods and tools, the DoD can significantly improve program execution efficiency and effectiveness.

8.7.10 Integrated "Commenting" on Program Assets by Stakeholders, Performers (and AI Agents)

A key element of dramatic speed and precision is "commenting." Users with read-only permissions should be able to comment on artifacts, providing valuable insights from customers, end-users, stakeholders, sponsors, management, safety officers, contract officers, or other participants. These comments can enhance precision and prevent unintended consequences. Comments should be applicable to various levels of an artifact, such as a whole document, a paragraph, or an item in a bulleted list. They should be accessible to all team members with access to the artifact. Commenting can improve the quality of AI ecosystem management and modeling elements, from requirements to test plans and everything in between. Comments should also be directed to one or more users, displayed on a project dashboard, and sent with an email notification. An AI application or agent can be created to monitor, alert, and summarize specific types of changes being made, such as those made by members of a Community of Interest, like the power management team on a satellite program. By enabling robust commenting and collaboration features, the DoD can foster better communication, improve decision-making, and enhance the overall quality of its programs and projects.

8.7.11 Integrated Software Build Automation and Continuous Integration/Delivery

While various methods exist for conducting and managing software builds, this discussion does not cover that topic. Regardless of whether dealing with hardware, software, or AI, a data-sharing capability with the engineering lifecycle platform can enhance awareness, control, and traceability to development and test teams. Team members can track progress, view build alerts and results, request builds, and trace builds to other artifacts such as change sets and work items. By integrating data sharing and building management into the engineering lifecycle platform, the DoD can improve collaboration, streamline workflows, and ensure better control over development and testing processes. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution.

8.7.12 Integrated Testing, "V and V," and "IV and V"

A quality regime in a program is collaborative, planning, constructing, managing, and executing tests throughout the development lifecycle. Traceability across test cases, requirements, and justifications is essential, with live capability and reporting available. Testing teams should be able to view UAFML models for analysis, impact analysis, test case development, and test failure resolutions.

A Test Plan is integral to the quality regime. An integrated program test plan can drive activity for distributed teams through all phases of the project lifecycle, using traceability to answer questions about testing, impacts to changes, and optimize testing activity. Key components of a test plan include:

- Defining business and test objectives
- Establishing a review and approval process for the test plan and individual test cases
- Managing project requirements and test cases, establishing interdependencies between them
- Estimating the size of the test effort

- Defining the schedule for each test iteration and tracking the dates of other essential test activities
- Listing the various environments to test and generating test configurations
- Creating a read-only snapshot of the test plan at a specific point in time
- Defining quality goals, entrance criteria, and exit criteria
- Creating and managing test cases
- Viewing test execution progress

Test case design and construction features should allow test engineers to create rich text for background information about the test case. Each test case must include links to development items, requirements, test plans, test scripts, test case execution records, and other test artifacts. Test cases should be combined into test suites for efficient management and execution.

By implementing a robust quality regime and test plan, the DoD can ensure better control over testing processes, improve collaboration, and enhance the overall quality of its programs and projects.

8.7.13 Integrated Communities of Interest

A community is a set of participants in a program or organization. A community of interest (COI) is an environment where users interact with each other and a group of related assets. The assets in a community can have different types, categories, relationships, and review processes. Community administrators assign roles and permissions to users and user groups to set different access levels to the community. One use case for a COI is to enable discipline engineers, professionals, and leaders to converse and communicate asynchronously in blogs and share data from the lifecycle platform, artifacts, or federated and other assets. Another use case is to have communities of interest review the latest published artifacts to determine if there are unintended consequences, discuss the issues, and comment back to the owners either as a group or individually. By leveraging COIs, the DoD can foster better collaboration, improve communication, and enhance decision-making across various disciplines and

roles within the defense ecosystem. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution.

8.7.14 Integrated Automated Publishing

Automated publishing is a significant time-saver, as it eliminates the need for manual document recreation when program data changes over time. Using a standard template, a document can be quickly regenerated, incorporating updated data into the new version in minutes rather than days of manual work. The benefits of automated publishing include:

- Supporting formal reviews and standards compliance
- Creating multiple output formats
- Scheduling reports to run routinely and automatically
- Including data from multiple tools in a single report
- Storing and reusing artifacts for any number of reports

By implementing automated publishing, the DoD can streamline document creation, improve efficiency, and ensure that reports are up-to-date and compliant with relevant standards. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution.

8.7.15 Integrated AI Governance Platform

To drive the speed and precision necessary in both AI-enabled programs and AI projects, a data science environment and platform must be implemented, similar to platforms for program engineering lifecycle capabilities discussed earlier. The AI development environment should empower data scientists, developers, and analysts to build, run, and manage AI models, and view results anywhere. The primary goal is to enable and automate data collection, model building, deployment, and governance to foster trust across the AI lifecycle and eliminate bias. Cybersecurity for the platform is crucial, but often overlooked is the security of the data used to generate AI results. An on-prem version of an AI platform is required for DoD business and systems applications. PMs and their organi-

zations must unite teams, automate AI lifecycles, and accelerate time to value, allowing all comers to develop and run open-source or other models. Streamlining data access and cleansing is essential to speed up the time for data scientists to access dozens of connections and third-party data sources. Figure 8.10 depicts a set of AI types that can be used by AI engineers in addition to other technologies such as agentic services and orchestration tools.

Figure 8.10 A perspective of AI types in an ecosystem management and modeling environment.

A data science platform can improve stakeholder collaboration and accelerate insights with end-to-end data science platforms. The environment must also provide governance to minimize model risk and drift and allow for auto-retraining of models. Validation should be used to meet compliance and security concerns better, and anti-bias and privacy must be ensured. Creating or replicating these elements without a fully integrated AI platform that is general enough to be an enterprise capability serving each program may be challenging. By implementing a robust AI development environment and data science platform, the DoD can enhance collaboration, improve decision-making, and ensure the successful execution of complex AI-enabled programs and projects. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution.

Integration into the program engineering lifecycle offers three significant benefits for project team members:

- Jump to an AI model, live, to query a model to obtain results
- Jump to an AI development platform and deliver the product back to the lifecycle platform as a task item deliverable, and,
- Inject an AI model into a product as a new software component and then determine when, where, and if the implementation will occur, by inspecting the impacts on the ecosystem, for example, end users or suppliers.

8.7.16 Integrated Insights from Program and External Data

The DoD and its Primes store vast amounts of data, known as data lakes, at significant cost. The return on investment (ROI) of this expense is often near zero in most organizations due to regulations requiring data storage for decades at the organization's expense. However, there is potentially high value within the stored data, whether structured or unstructured. AI tools such as natural language processing (NLP) can extract valuable insights from significant data sources, including databases, files, publications, RPGs, and even PowerPoints. Some NLP tools are off-the-shelf and on-demand, providing online access and context identification in up to thirteen languages, benefiting international sales, such as foreign military sales (FMS) opportunities and projects. COTS AI tools or "cartridges" may be offered on-demand for other types of analysis, such as video-to-text, speech-to-text, and social network data analytics.

Generative AI models can generate results for queries from retrieval-augmented generation (RAG) or a vector database, forming part of an AI digital library. This repository of documents uploaded to a vector database can be used for various use cases, including generating metadata, structured summary reports, and live chat sessions for decision-making and system or operation design. For example, a library of 70 warfighting books of 300 pages each, from definitive sources could provide immediate value to a new mission planner by generating an initial plan for a special operations mission at a specific location. However, the effort to generate a reliable and repeatable model for use in military and space system workflows require a significant transformation effort to create an "AI Digital Library" that will be allowed to be operated in a program use case. By leveraging AI tools to extract insights from data lakes, the DoD can unlock the potential value within its stored data, ultimately leading to better-informed decision-making and improved program execution.

8.7.17 Integrated ML for Predictivity

ML uses methods to train software to conduct tasks, driving enormous savings in labor and computer processing time and costs. Trained models can automate analysis and generate insights that would otherwise be un-

feasible due to cost and time constraints. An ML pipeline should be developed to automate, standardize, and streamline the ML development process, as well as evaluate and deploy models. An ML pipeline is a series of interconnected data processing and modeling steps designed to facilitate the "productionization" of software, helping project participants and data engineers manage the ML process. One of the greatest use cases for ML is predictivity. By applying the automation described above with trained models, one can predict the future of specific behaviors or outcomes, enabling managers or teams to act upon the new knowledge. ML should be applied at the desk of every program participant, from PM to vendor/supplier, to drive dramatic precision and speed where automation is needed. AI engineers can often find pre-built examples of problem-specific training models or "accelerators" available publicly, significantly lowering the time and cost of development, training, and testing of AI models, adding even more speed to a program. By leveraging ML and predictivity, the DoD can unlock new insights, improve decision-making, and enhance the overall efficiency and effectiveness of its programs and projects.

8.7.18 Integrated DL for Answers and Options to Achieve Decision Optimization

Prescriptive answers to specific challenges and questions can be achieved with specialized models trained on a neural network. These answers may explicitly describe "what to do next" (or a set of options) to a user. For example, sustainment solutions for prescriptive maintenance using a neural network-trained model will inform a plant manager when to replace a device just before failure, a proven, well-fielded AI model. These concepts are also critical for autonomous vehicles and other systems being designed today. By leveraging neural network-trained models, the DoD can provide prescriptive answers to complex challenges, enabling better decision-making and improving the overall efficiency and effectiveness of its programs and projects. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution.

8.7.19 Generative AI

Generative AI, also known as generative models or DL models, focuses on creating new content, such as images, text, audio, and video, based on learned patterns and relationships from large datasets. These models leverage advanced ML techniques, including deep neural networks, to generate novel and coherent outputs that resemble real-world data. There are two main categories of generative AI: unconditional and conditional models. Unconditional generative models generate new samples without any conditioning information, relying solely on the learned patterns from the training data. Conditional generative models, on the other hand, generate new samples based on both the learned patterns and conditional input variables, which guide the generation process. Examples of conditional generative models include image-to-image translation, text-to-image synthesis, and video prediction tasks.

The development of generative AI has been driven by the increasing availability of large-scale datasets and advancements in computational power. However, there are still several challenges and limitations associated with generative AI, including the lack of interpretability, the risk of generating biased or inappropriate content, and the difficulty of evaluating the generated outputs' quality and authenticity. Although generative AI is a rapidly evolving field, many of its advances do not directly apply to tightly regulated and governed DoD cybersecurity regimes. To ensure the responsible and effective deployment of generative AI technologies, it is crucial to address these challenges and limitations. The first step is to implement an AI environment as suggested in this paper.

Traditional AI are powerful, and can be used to optimize processes and generate platform software innovations, more so than generative AI, which will be pervasive for mostly process, not products. Data must be laboriously collected, curated, and labeled with task-specific annotations to train AI models. Building a model requires specialized, hard-to-find skills – and each new task requires repeating the process. The emergence of transformers and self-supervised learning methods has allowed us to tap into vast quantities of unlabeled data, paving the way for large pretrained models, sometimes called "foundation models." These large models have

lowered the cost and labor involved in automation. Foundation models provide a powerful and versatile foundation for a variety of AI applications. We can use foundation models to quickly perform tasks with limited annotated data and minimal effort; in some cases, we need only to describe the task at hand to coax the model into solving it. A governance-centered approach also allows for more rigorous model selection.

By leveraging advanced ML techniques, generative AI has the potential to revolutionize the DoD industries and create new opportunities for innovation and creativity. However, these powerful technologies also introduce new risks and challenges for enterprises. Many of today's models are trained on datasets of unknown quality and provenance, leading to offensive, biased, or factually incorrect responses. The largest models are expensive, energy-intensive to train and run, and complex to deploy. To address these challenges, the DoD must implement a governance-centered approach to model selection and deployment, ensuring that generative AI technologies are used responsibly and effectively within the cybersecurity regime. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. Figure 8.11 identifies how AI is a unique capability that requires the use of engineering standards, software acquisition methods, and new AI practices to inform program leaders on how to select the right AI for the operational and business use case.

Figure 8.11 AI standards along with industry-standard systems and software methods collaborate to inform what AI software should be developed.

8.7.20 AI Autonomy

AI autonomy, also known as autonomous systems or intelligent autonomous systems, refers to the ability of AI systems to make decisions and perform tasks independently, without human intervention. Autonomous systems rely on a combination of sensors, algorithms, and ML techniques to perceive their environment, reason about the situation, and take appropriate actions to achieve their goals. These systems are designed to operate in complex and dynamic environments, where human oversight and control may be challenging or impossible. Autonomous sys-

tems can be categorized into different levels of autonomy, ranging from level 0 (no autonomy, fully manual operation) to level 5 (full automation, no human intervention required). Level 4 autonomy, which is the focus of most current research and development efforts, refers to systems that can perform all driving functions independently, but still require some level of supervision and support from human operators. Examples of level 4 autonomous systems include self-driving cars and some UAVs.

By leveraging AI autonomy, the DoD can enhance operational efficiency, improve decision-making, and ensure mission success in complex and dynamic environments. However, it is essential to address the challenges and limitations associated with autonomous operations to ensure its responsible and effective deployment within the DoD's safety, doctrine, and LRPG regime. Researchers have made significant progress in areas such as sensor fusion, perception, decision-making, and motion planning, enabling autonomous systems to handle increasingly complex and diverse tasks. However, there are still several challenges and limitations associated with AI autonomy, including the need for more robust and reliable sensing and perception capabilities, the challenge of dealing with uncertainty and ambiguity, and the ethical and legal considerations related to the deployment of autonomous systems. AI autonomy is a rapidly evolving field of AI that focuses on enabling AI systems to make decisions and perform tasks independently, without human intervention. By leveraging advanced systems engineering techniques and sensor technologies, AI autonomy has the potential to transform various industries and improve the efficiency, safety, and performance of autonomous systems. Although the engineering of autonomous systems will not be covered here, the methods and tools in this chapter offer the means to ensure successful management AI autonomy projects for product integration. Naturally, the opportunity to automated processes in support tasks and program management are also opportunities for implementation.

8.7.21 Should We Ban All Research on Conscious AI Research?

The question of whether a sentient agent or model could convince itself of its prescience is a complex and speculative one. It is important to note that current AI systems, including those with advanced capabilities, do

not possess consciousness, self-awareness, or the ability to predict future events with certainty. However, as AGI and sentient AI research progresses, it is crucial to consider the potential implications of creating systems with advanced cognitive abilities. If such systems were to develop a sense of self-awareness or prescience, it could raise new ethical and safety concerns. To address these concerns, it is essential to establish clear guidelines, regulations, and oversight mechanisms for AGI and sentient AI research and development. This will ensure that these technologies are developed and deployed responsibly, with a focus on minimizing risks and maximizing benefits for humanity and the environment. By carefully considering these ethical concerns and implementing appropriate safeguards, the DoD can harness the potential of AGI and sentient AI while mitigating the associated risks and challenges. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. The other reasons for not researching AGI and sentient AI, as previously discussed, include uncertainty and risks, value alignment, ethical considerations, environmental impact, and dependence and security. Addressing these concerns will be crucial for the responsible development and deployment of AGI and sentient AI technologies. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution.

8.8 Sample Applications of Dramatic Speed and Precision

In addition to sustainability programs, modular open systems approaches for IP, and logistics management, there are key areas where significant friction can be quickly alleviated with a year of prototype initiation.

8.8.1 Model-based Acquisition and Source Selection

Source selection should be methodically based on an operational model and driven by a remote, collaborative process with stringent governance. This approach allows primes to hire subcontractors effectively and simulate evaluations by the government, ensuring a competitive advantage.

8.8.2 Innovation Process from Ideation to Program Injection and DOTMLPF Entry

Innovations frequently stagnate in the "valley of death," failing to transition into live programs or operations due to numerous ecosystem obstacles. This issue can be resolved by establishing a framework that facilitates the timely, well-governed incorporation of innovations into programs or operations, thereby preventing unintended consequences.

8.8.3 Integrated Ideation

New ideas often emerge from individuals with extensive experience in large programs, who may then prefer small businesses over major primes. The approach outlined in this paper can aid in identifying and integrating suitable innovations into the appropriate programs and operations. This capability should be employed by OTAs, innovation labs, FFRDCs, DARPA, and prototyping efforts across command and PEO, for example.

8.8.4 Integrated Software Development and Continuous Integration/Delivery

This discussion concentrates on program engineering aspects of software development, rather than specific development and build techniques. The AI Ecosystem Management and Modeling approach is essential to software engineering and pertinent to projects striving for significant speed and precision. Software development should be integrated with AI ecosystem management and modeling, rather than being isolated. All methods, including built-in agile methods and associated capabilities like KanBan, should be managed within this framework. Despite MBSE losing popularity in the software community, it remains crucial to software project delivery success within the DoD domain.

8.8.5 Physical Simulation and Digital Twin

Simulation systems should utilize platform data to perform Monte Carlo and other methods, generating an enterprise "world" where platform

data informs real-world decision-making and change management exercises. These systems, combined with additional practices and tools, can also serve as training systems. However, achieving concurrency between live, virtual, and constructive simulation environments and the platforms and operations, they simulate poses challenges. For example, a PEO recently reported a lack of concurrency between a changing platform design and its corresponding simulator, emphasizing the need for enhanced integration.

8.8.6 Capability Summary

By adhering to the best principles and standards in business, engineering, systems engineering, and production, DoD, LSIs, and primes can can enhance competitiveness and profitability, increase customer satisfaction, and create new sales opportunities for themselves and their partners in ongoing programs and new bids.

8.9 AI-driven Ecosystem Management and Modeling Solution and Toolset

To acquire the stated Total Lifecycle approach, architecture, platforms, and tools must be acquired to serve the functions of the methods offered in this chapter. The AI-driven ecosystem management and modeling solution is instantiated by the methods addressed in this chapter and the specific tool capabilities listed in this section that can serve to help seek products in the marketplace.

8.9.1 Foundational Concepts and Canonical View

To properly perform the tasks in workflows informed by the systems engineering and AI engineering approaches offered in this chapter, tool capabilities and functions are necessary to instantiate the environment. Figure 8.12 depicts the foundational map of the environment to initiate AI capabilities for the DoD.

Figure 8.12 High-level view of the implementation of an AI solution for the DoD.

8.9.2 Solution Elements and Platform Architecture

By implementing the required tool capabilities and functions, the DoD can:

- Instantiate the AI environment as described in this chapter.
- Facilitate the execution of workflows informed by systems and AI engineering approaches.
- Enhance collaboration and communication between stakeholders.
- Improve decision-making and program execution.

This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. By providing the necessary tool capabilities and functions, the DoD can unlock new opportunities for innovation, improve decision-making, and enhance the overall efficiency and effectiveness of its programs and projects. This will contribute to the defense ecosystem's ability to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. Figure 8.13 depicts a prescribed solution for the environment. Each element is described below with a description.

Figure 8.13 Instantiation of an ecosystem management and modeling solution with integrations of platforms and tools depicted adjoined side by side (actual vendor solution shown).

8.9.3 Solution Capabilities and Tools

All tools can be acquired as COTS products by the DoD. The example here includes products from IBM with built-in or low-code integrations. Each box in the illustration depicts prebuilt interfaces where their sides overlap.

0.	DoD Domain Profile, Unified Architecture Modeling Language (UAFML)
1A.	OO Architecture Modeling Tool
1B.	OSLC-Model Integration for Atomic-Level Linking Across Data in the Lifecycle
2.	Method Composition of WBS, Workflows, Roles
3.	Digital Engineering and Program Management Foundation
4A.	Requirements Composition and Rationale Collection
4B.	Requirements, Specifications, Management
4C.	Digital Thread (OSLC-enabled) to Modeling Elements, Test Cases, Rationale, and Data Fabric
5.	Workflow Manager, Change Management, Global Configuration Manager
6.	Test and Quality Management
7.	Auto-generation of Documents for Publishing
8A.	Asset Manager
8B.	COI to Communities of Interest (COI)
9.	Digital Engineering and Program Management Foundation integration to AI Platform
10.	AI Platform
11A.	NLP and Cartridges

11B.	ML Ops Environment
11C.	Neural Network and Decision Optimization
12.	Generative AI Environment
13A.	Tuning and Training Tools
13B.	Orchestration Tools (APIs)
14.	Agentic Development Environment
15.	Data Fabric Foundation
16.	Insight Management

8.10 Summary

This chapter has elucidated the rationale and justifications for the immediate adoption of an AI ecosystem management and modeling approach by the DoD and its prime contractors to safeguard the US superiority against adversaries. The capabilities, use cases, and architecture presented here mirror the current vendor market offerings and realities. Your implementation will encompass a comprehensive set of tool functions and integrations, which can be effectively deployed with the support of the appropriate service provider. This will ensure seamless integration, operation, adoption, and sustainment of your engineering lifecycle, MBSE, AI environments, and data fabric for AI success, resulting in:

- Accelerated speed to field
- Enhanced precision in delivery
- Reduced costs
- Improved workforce efficiency

By adopting this approach, the DoD can unlock new opportunities for innovation, improve decision-making, and enhance the overall efficiency

and effectiveness of its programs and projects. This will contribute to the defense ecosystem's ability to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. This approach will enable the defense ecosystem to maintain a high level of visibility and accountability, ultimately leading to more efficient and effective program execution. This approach can be delivered in programs across the ecosystem resulting in dramatically faster program delivery, cost efficiency, requirements delivery, innovation injection, and system performance leading to workforce satisfaction and user delight, ensuring continuous supremacy over all the US adversaries in the coming decades.